

11 October 2024

PHY201 Exam on chapter(s): 9, 10

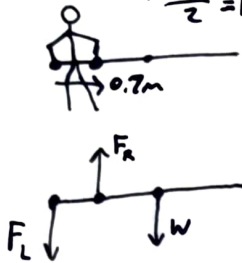
Name: Solutions

These questions assess your understanding of the material from Chapters 9, 10 of OpenStax College Physics. Please write your name on the sheets of paper that you use and show all your work.

1. Ryan, a construction worker, holds up a plank of wood 3.7-m long. His left hand pushes an end of the plank down with a force F_L , while his right hand pushes the plank up with a force F_R at a distance of 0.7 m from the left hand. The plank is held in equilibrium. The plank has a mass of 8 kg and its center of gravity is located at the middle of the plank. Calculate the magnitude of force F_R .

$\frac{3.7\text{m}}{2} = 1.85\text{m}$ $\therefore a = 0 \therefore F_{\text{net}} = 0$
 $\alpha = 0 \therefore \tau_{\text{net}} = 0$

ANSWER: 207.2 N



$F_{\text{net},y} = 0$
 $F_R - F_L - W = 0$

$\tau_{\text{net}} = 0$
 $\tau_R - \tau_W = 0$
 $\tau_R = \tau_W$
 $r_{\perp R} F_R = r_{\perp W} W$
 $F_R = \frac{r_{\perp W}}{r_{\perp R}} Mg = \frac{1.85\text{m}}{0.7\text{m}} (8\text{kg})(9.8\text{m/s}^2) = 207.2\text{N}$

lay pivot point on left end

2. Scott, a deep-sea fisherman, hooks a big fish that swims away from the boat, pulling the fishing line from the fishing reel. The whole system is initially at rest and the fishing line unwinds from the reel at a radius of 5.1 cm from the reel's axis of rotation. The reel is given an angular acceleration of 139 rad/s^2 for 2.8 s. How many meters of fishing line come off the reel in this time?

ANSWER: 27.8 m

find Δx
 $\Delta\theta = \frac{\Delta s}{r} = \frac{\Delta x}{r}$

(note $5.1\text{cm} \left| \frac{1\text{m}}{100\text{cm}} \right| = 5.1 \times 10^{-2}\text{m}$)

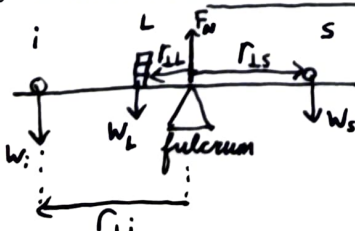
cm is not the SI unit

$\Delta x = r \Delta\theta$
 $\Delta x = r \left(\omega_0 t + \frac{1}{2} \alpha t^2 \right)$ ($\therefore \Delta\theta(t) = \omega_0 t + \frac{1}{2} \alpha t^2$)
 $\Delta x = r \left(0 + \frac{1}{2} \alpha t^2 \right)$
 $\Delta x = \frac{\alpha t^2 r}{2} = \frac{(139\text{ rad/s}^2)(2.8\text{s})^2(5.1 \times 10^{-2}\text{m})}{2}$

$\Delta x = 27.8\text{m}$

3. Two children, Stephen and Ian, and a stack of lego bricks are balancing on a long seesaw of negligible mass. Stephen has a mass of 24 kg and sits 3.7 m away from the fulcrum. Ian, who has a mass of 17 kg, sits on the other side of the fulcrum, opposite to Stephen. The stack of lego bricks has a mass of 2.2 kg and is on Ian's side of the seesaw, a distance of 0.5 m away from the fulcrum. What is Ian's distance from the fulcrum?

ANSWER: 5.159 m



balancing $\therefore a = 0 \therefore F_{\text{net}} = 0$
 $\therefore \alpha = 0 \therefore \tau_{\text{net}} = 0$

(choose fulcrum as pivot point)

$F_{\text{net}} = \Sigma F_y = 0$
 $F_N - W_i - W_L - W_S = 0$

$F_N = W_i + W_L + W_S$
 $F_N = (m_i + m_L + m_S)g$
 $F_N = (17 + 2.2 + 24\text{kg}) 9.8\text{m/s}^2$

$F_N = 423.4\text{N}$

$\tau_{\text{net}} = \Sigma \tau = 0$

$\tau_i + \tau_L + \tau_F + \tau_S = 0$

$r_{\perp i} m_i g + r_{\perp L} m_L g + r_{\perp F} F_N - r_{\perp S} m_S g = 0$

(divide all by g) (zero)

$r_{\perp i} m_i + r_{\perp L} m_L - r_{\perp S} m_S = 0$

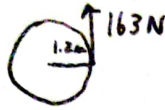
$r_{\perp i} = \frac{1}{m_i} (r_{\perp S} m_S - r_{\perp L} m_L)$ (cont. on left side)

$r_{\perp i} = \frac{1}{17\text{kg}} (3.7\text{m}(24\text{kg}) - 0.5\text{m}(2.2\text{kg}))$

$r_{\perp i} = 5.159\text{m}$

4. Samantha is with her daughter, Chloe, at a park. She is continuously pushing on a 58-kg merry-go-round with a force of 163 N at its edge, 1.2 m away from the center. Chloe watches excitedly. Calculate the angular acceleration that is produced in the merry-go-round when no one is on it. Consider the merry-go-round to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.

find α



$m = 58 \text{ kg}$

$\tau_{\text{net}} = I\alpha$
& $\tau_{\text{net}} = r_{\perp} F_{\text{net}}$

$\therefore I\alpha = r_{\perp} F_{\text{net}}$

$\alpha = \frac{1}{I} (r_{\perp} F_{\text{net}})$

$\alpha = \frac{1}{\frac{1}{2}MR^2} (R F_{\text{net}})$

$\alpha = \frac{2F_{\text{net}}}{MR} = \frac{2(163 \text{ N})}{(58 \text{ kg})(1.2 \text{ m})} = 4.68 \frac{\text{rad}}{\text{s}^2}$

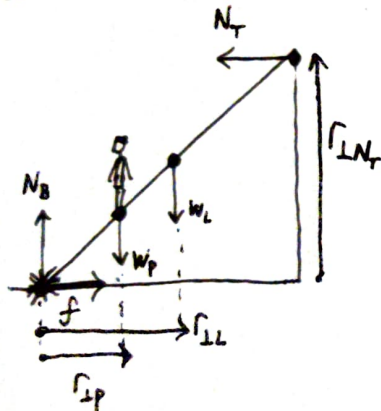
ANSWER: $4.68 \frac{\text{rad}}{\text{s}^2}$

units note

$\times \frac{\text{N}}{\text{kg}\cdot\text{m}} = \frac{\text{kg}\cdot\text{m}}{\text{s}^2\cdot\text{kg}\cdot\text{m}} = \frac{1}{\text{s}^2}$
& rad has no dimension
 $\therefore \frac{\text{N}}{\text{kg}\cdot\text{m}} = \frac{\text{rad}}{\text{s}^2}$

Sketch is very important here.
Say pivot point at bottom.

5. Nathan places a ladder of mass 19 kg against his house. The feet of the ladder are resting on his concrete patio at a distance of 5.3 m from the house. The top of the ladder rests against a plastic rain gutter. The friction between the ladder and gutter is negligible. The top of the ladder is at a height of 7.5 m above the concrete patio. The ladder's center of gravity is at its midpoint. Nathan is standing on a rung at a horizontal distance of 3.8 m from the house. If Nathan's mass is 74 kg, what is the magnitude of the friction force between the feet of the ladder and the concrete?



$N_T = ?$ normal force at top

$N_B = ?$ normal force at bottom

$f = ?$ find fric. force

$W_P = M_P g = (74 \text{ kg})(9.8 \frac{\text{m}}{\text{s}^2}) = 725.2 \text{ N}$

$W_L = M_L g = (19 \text{ kg})(9.8 \frac{\text{m}}{\text{s}^2}) = 186.2 \text{ N}$

$r_{\perp W_T} = 7.5 \text{ m}$

$r_{\perp L} = \frac{5.3 \text{ m}}{2} = 2.65 \text{ m}$

$r_{\perp P} = 5.3 \text{ m} - 3.8 \text{ m} = 1.5 \text{ m}$

ANSWER: 210.83 N

at equilibrium (static) $\therefore v=0$
 $a=0$
 $\alpha=0$

$\Sigma F_x = 0 \therefore a_x = 0$

$f - N_T = 0$

$f = N_T$

$\Sigma F_y = 0 \therefore a_y = 0$

$N_B - W_P - W_L = 0$

$N_B = (M_P + M_L)g$

$\Sigma \tau = 0 \therefore \alpha = 0$

$\vec{\tau}_{N_B} + \vec{\tau}_f + \vec{\tau}_{W_P} + \vec{\tau}_{W_L} + \vec{\tau}_{N_T} = 0$

$0 + 0 - r_{\perp P} M_P g - r_{\perp L} M_L g + r_{\perp W_T} N_T = 0$

$r_{\perp W_T} N_T = (r_{\perp P} M_P + r_{\perp L} M_L)g$

(continued on scratch)

6. Olivia places an outdoor sandwich-board advertising sign on the ground. The sign used to have a chain 0.52 m vertically below its hinge, but the chain is broken. The sign's mass is 8 kg. The sign has a vertical height of 1.69 m. The hinge is opened so that the feet of the sign measure 2.6 m apart. What is the minimum coefficient of static friction between the feet of the sign and the sidewalk? To draw the sign, you may draw a large A. The chain, now broken, would have resembled the horizontal line in the A and the hinge is at the top of the A.

pivot point bottom right.

consider right side only. Symmetric.

ANSWER: 0.3846

Equilibrium $\therefore a=0 \therefore F_{\text{net}}=0$
 $\therefore \alpha=0 \therefore \tau_{\text{net}}=0$

$\Sigma F_x = 0 \therefore a_x = 0$

$N_T - f_s = 0$

$N_T = f_s = \mu_{\text{min}} N_B$

$\Sigma F_y = 0 \therefore a_y = 0$

$N_B - W = 0$

$N_B = W = \frac{8 \text{ kg}(9.8 \frac{\text{m}}{\text{s}^2})}{2}$

$N_B = 39.2 \text{ N}$

$\Sigma \tau = 0 \therefore \alpha = 0$

$\vec{\tau}_B + \vec{\tau}_f + \vec{\tau}_W + \vec{\tau}_T = 0$

$0 + 0 + r_{\perp W} W - r_{\perp T} N_T = 0$

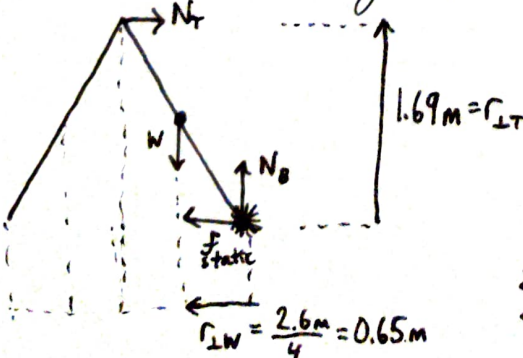
$r_{\perp W} W = r_{\perp T} N_T$

$N_T = \frac{1}{r_{\perp T}} (r_{\perp W} W) = \frac{1}{1.69 \text{ m}} (0.65 \text{ m} (\frac{8 \text{ kg}}{2} (9.8 \frac{\text{m}}{\text{s}^2}))) = 15.08 \text{ N}$

$\mu_{\text{min}} = \frac{N_T}{N_B}$

$\mu_{\text{min}} = \frac{15.08 \text{ N}}{39.2 \text{ N}}$

$\mu_{\text{min}} = 0.3846$



#5 continued

$$r_{\text{net}} N_T = (r_{\perp p} m_p + r_{\perp L} m_L) g$$

$$N_T = \frac{1}{7.5 \text{ m}} (1.5 \text{ m} (74 \text{ kg}) + 2.65 \text{ m} (19 \text{ kg})) 9.8 \text{ m/s}^2$$

$$N_T = 210.83 \text{ N} = 210.83 \frac{\text{kg} \cdot \text{m}}{\text{s}^2}$$

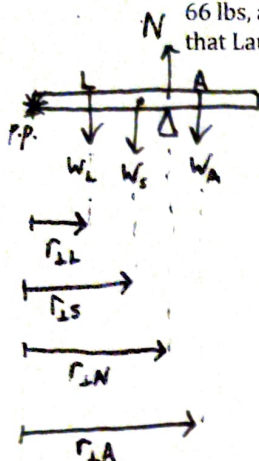
Recall that $f = N_T$ from $\Sigma F_x = 0$

$$\therefore \boxed{f = f_k = 210.83 \text{ N}}$$

5.6 ft 5.6 ft



7. Adam is building a seesaw for his daughter, Lauren, and plans on making it 14 ft long. He plans to sit on one side, 5.6 ft away from the center of the seesaw, and his daughter will sit on the other side, 5.6 ft away from the center of the seesaw. Adam has weighed himself, his daughter, and the seesaw and learned that his weight is 171 lbs, his daughter's weight is 66 lbs, and the weight of the seesaw is 12.9 lbs. He wants to place the fulcrum so that the seesaw is perfectly balanced, so that Lauren can cause a rotation in the seesaw just as easily as himself. Where must the fulcrum be placed?



$$W_A = 171 \text{ lb}$$

$$W_L = 66 \text{ lb}$$

$$W_S = 12.9 \text{ lb}$$

$$r_{LL} = \left(\frac{14}{2} - 5.6\right) \text{ ft} = 1.4 \text{ ft}$$

$$r_{LS} = 14 \text{ ft} / 2 = 7 \text{ ft}$$

$$r_{LN} = x \text{ (solve for this)}$$

$$r_{LA} = \left(\frac{14}{2} + 5.6\right) \text{ ft} = 12.6 \text{ ft}$$

equilibrium $\therefore a=0 \therefore F_{\text{net}}=0$
 $\alpha=0 \therefore \tau_{\text{net}}=0$

$$\sum F_y = 0 \therefore a_y = 0$$

$$N - W_L - W_S - W_A = 0$$

$$N = (M_L + M_S + M_A)g \text{ (with SI units)}$$

$$N = W_L + W_S + W_A$$

$$N = (66 + 12.9 + 171) \text{ lb} = 249.9 \text{ lb}$$

ANSWER: 2.353 ft from center, on Adam's side

$$\sum \tau = 0 \therefore \alpha = 0$$

$$\vec{r}_L + \vec{r}_S + \vec{r}_N + \vec{r}_A = 0$$

$$-r_{LL}W_L - r_{LS}W_S + r_{LN}N - r_{LA}W_A = 0$$

$$r_{LN}N = r_{LL}W_L + r_{LS}W_S + r_{LA}W_A$$

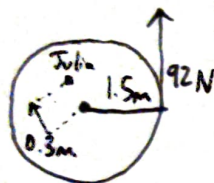
$$\text{Solve for } r_{LN} \text{ or "x"}$$

$$r_{LN} = \frac{1}{N} (r_{LL}W_L + r_{LS}W_S + r_{LA}W_A)$$

$$r_{LN} = \frac{1}{249.9 \text{ lb}} (1.4 \cdot 66 + 7 \cdot 12.9 + 12.6 \cdot 171)$$

$$r_{LN} = 9.353 \text{ ft (from left side)}$$

8. Jessica is with her daughter, Julia, at a park. Julia has climbed onto a merry-go-round. Jessica is continuously pushing on the 46-kg merry-go-round with a force of 92 N at its edge, 1.5 m away from the center. Calculate the angular acceleration that is produced in the merry-go-round while 21-kg Julia is 0.3 m away from its center. Consider the merry-go-round to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.



$$\tau_{\text{net}} = I\alpha \text{ --- (N2 Law rot.)}$$

$$\& \tau_{\text{net}} = r_{\perp} F_{\text{net}} \text{ --- (definition)}$$

$$\therefore r_{\perp} F_{\text{net}} = I\alpha$$

$$\alpha = \frac{1}{I} (r_{\perp} F_{\text{net}}) = \frac{R F_{\text{net}}}{(\frac{1}{2}MR^2 + M_j R_j^2)} = \frac{1.5 \text{ m} (92 \frac{\text{kgm}}{\text{s}^2})}{(\frac{1}{2}(46 \text{ kg})(1.5 \text{ m})^2 + 21 \text{ kg}(0.3 \text{ m})^2)} = 2.573 \frac{\text{rad}}{\text{s}^2}$$

ANSWER: 2.573 rad/s²

$$M = 46 \text{ kg}$$

$$R = 1.5 \text{ m}$$

$$M_j = 21 \text{ kg}$$

$$R_j = 0.3 \text{ m}$$

Why is $I_{\text{system}} = \frac{1}{2}MR^2 + M_j R_j^2$?
 Use the parallel-axis theorem.
 $I_{\text{system}} = I_{\text{cylinder}} + I_{\text{point mass}}$

9. Ethan, a deep-sea fisherman, hooks a big fish that swims away from the boat, pulling the fishing line from the fishing reel. The whole system is initially at rest and the fishing line unwinds from the reel at a radius of 3.9 cm from the reel's axis of rotation. The reel is given an angular acceleration of 60 rad/s² for 2.3 s. At what speed is the fishing line leaving the reel after 2.3 s elapses?

ANSWER: 5.382 m/s

$$\text{find } v(t=2.3 \text{ s})$$

$$v_i = 0$$

$$\omega_0 = 0$$

$$R = 3.9 \times 10^{-2} \text{ m} = 0.039 \text{ m}$$

$$\alpha = 60 \frac{\text{rad}}{\text{s}^2}$$

$$v = R\omega \therefore v(t) = R\omega(t)$$

$$v(t) = R(\omega_0 + \alpha t)$$

$$v(t) = R\omega_0 + R\alpha t$$

$$v(t=2.3 \text{ s}) = (0.039 \text{ m})(0 \frac{\text{rad}}{\text{s}}) + (0.039 \text{ m})(60 \frac{\text{rad}}{\text{s}^2})(2.3 \text{ s})$$

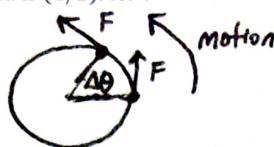
$$v(t=2.3 \text{ s}) = 5.382 \frac{\text{m} \cdot \text{rad} \cdot \text{s}}{\text{s}^2} = 5.382 \frac{\text{m}}{\text{s}}$$

$$\omega(t) = \omega_0 + \alpha t$$

Note:
there are other ways to do this.
 $W_{\text{net}} = \Delta KE = T_{\text{net}} \Delta \theta$
is another path

what is α ?

$$\begin{aligned} T_{\text{net}} &= I\alpha \\ \& \quad T_{\text{net}} &= r_{\perp} F_{\text{net}} \\ \therefore I\alpha &= r_{\perp} F_{\text{net}} \\ \alpha &= \frac{RF}{I} = \frac{2RF}{MR^2} = \frac{2F}{MR} \end{aligned}$$



$$\begin{aligned} F &= 302 \text{ N} \\ \Delta \theta &= 1.2 \text{ rad} \\ R &= 0.55 \text{ m} \\ M &= 74 \text{ kg} \\ \omega_0 &= 0 \frac{\text{rad}}{\text{s}} \end{aligned}$$

ANSWER: 199.32 J

find KE_f

$$KE_f = \frac{1}{2} I \omega_f^2$$

$$KE_f = \frac{1}{2} \left(\frac{1}{2} MR^2 \right) (\omega_0^2 + 2\alpha \Delta \theta)$$

$$KE_f = \frac{1}{4} MR^2 (2\alpha \Delta \theta) = \frac{MR^2 \alpha \Delta \theta}{2}$$

$$KE_f = \frac{MR^2 \left(\frac{2F}{MR} \right) \Delta \theta}{2} = RF \Delta \theta$$

$$KE_f = (0.55 \text{ m})(302 \text{ N})(1.2 \text{ rad}) = \boxed{199.32 \text{ J}}$$

11. A typical small rescue helicopter has four blades, each of which is 4.1-m long and has a mass of 47 kg. The blades can be approximated as thin rods that rotate about one end of an axis perpendicular to their length, so that the moment of inertia of a blade is $(1/3)ML^2$. The helicopter has a total loaded mass of 1270 kg. Calculate the rotational kinetic energy that the helicopter possesses when its blades rotate at 275 rpm.

$$\begin{aligned} \omega_b &= 275 \frac{\text{rev}}{\text{min}} \\ \text{or } 28.798 \frac{\text{rad}}{\text{s}} \end{aligned}$$

$$M_{\text{blade}} = 47 \text{ kg}$$

$$L_{\text{blade}} = 4.1 \text{ m}$$

$$I_{\text{blade}} = \frac{1}{3} M_b L_b^2$$

The helicopter has four blades, each of which contributes rotational kinetic energy to the whole system.

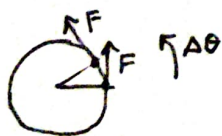
$$\text{i.e. } KE_{\text{rot}} = 4 KE_{\text{rot, blade}} = 4 \left(\frac{1}{2} I_b \omega_b^2 \right) = 2 I_b \omega_b^2$$

$$KE_{\text{rot}} = 2 \left(\frac{1}{3} M_b L_b^2 \right) \omega_b^2 = \frac{2}{3} (47 \text{ kg})(4.1 \text{ m})^2 (28.798 \frac{\text{rad}}{\text{s}})^2$$

$$KE_{\text{rot}} = 436816 \text{ J or } \boxed{436.8 \text{ kJ}}$$

ANSWER: 436.8 kJ

12. Charles is spinning a large grindstone by placing his hand on its edge and exerting a force through part of a revolution. How much work does he do if he exerts a force of 138 N through a rotation of 0.34 rad (19°)? He exerts the force perpendicular to the grindstone's 38-cm radius and the effects of friction are negligible. Consider the grindstone to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.



$$\begin{aligned} F &= 138 \text{ N} \\ \Delta \theta &= 0.34 \text{ rad} \\ R &= 0.38 \text{ m} \end{aligned}$$

ANSWER: 17.8 J

Two paths you can take:

Path 1: $W_{\text{net}} = T_{\text{net}} \Delta \theta$

$$W_{\text{net}} = r_{\perp} F_{\text{net}} \Delta \theta$$

$$W_{\text{net}} = RF \Delta \theta$$

$$W_{\text{net}} = (0.38 \text{ m})(138 \text{ N})(0.34 \text{ rad})$$

$$W_{\text{net}} = \boxed{17.8 \text{ J}}$$

Path 2: $W_{\text{net}} = \Delta KE = KE_f - KE_0$

$$W_{\text{net}} = \frac{1}{2} I \omega_f^2$$

$$W_{\text{net}} = \frac{1}{2} \left(\frac{1}{2} MR^2 \right) (\omega_0^2 + 2\alpha \Delta \theta)$$

$$W_{\text{net}} = \frac{1}{4} MR^2 \left(2 \left(\frac{RF}{I} \right) \Delta \theta \right)$$

$$W_{\text{net}} = \frac{1}{2} MR^2 \left(\frac{RF}{\frac{1}{2} MR^2} \right) \Delta \theta$$

$$W_{\text{net}} = RF \Delta \theta$$

$$W_{\text{net}} = (0.38 \text{ m})(138 \text{ N})(0.34 \text{ rad})$$

$$W_{\text{net}} = \boxed{17.8 \text{ J}}$$

$$\begin{aligned} T_{\text{net}} &= I\alpha = r_{\perp} F_{\text{net}} \\ \therefore \alpha &= \frac{RF}{I} \end{aligned}$$

PHY201 Exam on chapter(s): 9, 10

1. 207.2 N
2. 27.79 m
3. 5.159 m
4. 4.68 rad/s^2
5. 210.83 N
6. 0.38
7. 2.4 ft from the center, on Adam's side
8. 2.57 rad/s^2
9. 5.38 m/s
10. 199.32 J
11. 436.8 kJ
12. 17.8 J